Cameron Souza

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EDUCATION

Ph.D., Environmental Sciences

Oregon State University, Corvallis, Oregon · Quantitative Analysis track · matriculating Fall 2026

- Advisor: Dr. Carolyn Fonyo.
- Dissertation: *A Hierarchical Bayesian State-Space Framework for Boundary-Flux Modulation of Community Function: Theoretical Foundations for Bioregenerative Closed Systems, Empirically Validated on the Hawaiian Archipelago.*
- Approved graduate transfer credit determination received September 2026.

M.A., Data Science

Harvard University · Magna Cum Laude

B.S., Marine Biology and Sustainable Aquaculture

Unity College (now Unity Environmental University) · Magna Cum Laude

- Earth science coursework included volcanology and oceanography.

TEACHING EXPERIENCE

Adjunct Faculty, Distance Education

Unity Environmental University · 2026 to present

- Instructor of record for MATH 301, Statistical Modeling for Life and Environmental Sciences, a five-week fully online section.
- Five modules: descriptive statistics and study design, linear regression and correlation, multiple linear regression, logistic regression, Poisson regression.
- All student assignments are completed in R. Course is delivered and graded in Canvas.

Teaching Assistant and Grader, Oceanography

Unity College · One semester

- Graded undergraduate coursework and supported instruction in oceanography.

Applied Instruction and Tutoring

Various, ongoing

- Statistics and R tutoring, individual and small group.
- Scientific diving instruction and dive safety briefings.
- Vessel operation training for crew and owners under a USCG Master license.

CURRICULUM DEVELOPMENT

Co-developer, Technical Skills Curriculum

With a Unity Environmental University faculty colleague

- Co-authored R and statistics curriculum and Python data science modules.
- Built Canvas course shells, module structure, assignment rubrics, and gradebook configuration.
- Some sections have been delivered. The remainder are built and pending delivery.

RESEARCH EXPERIENCE

Doctoral Research

Oregon State University · 2026 to present

- Hierarchical Bayesian multi-source data fusion for bounded biogeochemical systems, with explicit uncertainty propagation.
- Empirical system is the Hawaiian Archipelago. Work treats volcanic island weathering and basalt-derived nutrient flux as the driving boundary process.
- Primarily satellite remote sensing and federal and public environmental datasets.

Field Research

Center for Marine Debris Research, Oʻahu, Hawaiʻi

- Shoreline transects and beach sediment sampling along basalt coastline.

Field Research

Coral Restoration Foundation, Florida Keys

- Benthic and carbonate substrate assessment and reef framework monitoring.

TECHNICAL SKILLS

- **Statistical computing.** R, Python. Hierarchical Bayesian inference, state-space modeling, uncertainty propagation.
- **Regression methods.** Linear, multiple linear, logistic, and Poisson regression. Study design and descriptive statistics.
- **Geospatial and remote sensing.** Satellite data products and federal and public environmental datasets.
- **Instructional platforms.** Canvas LMS, including shell build, rubric design, and gradebook configuration.

LICENSES AND CERTIFICATIONS

- USCG 100-Ton Master Captain.
- AAUS Scientific Diver.
- PADI Rescue Diver.

PROFESSIONAL EXPERIENCE

- **Founder, Terra Laboratories.** San Diego. Bioregenerative hardware and inference infrastructure.
- **Owner and operator, Solarium Marketing.** Data science driven marketing for outdoor and earth-based brands.
- **Technical and marine services, Watermark Yachts.** San Diego.
- **Operator, Mission Bay Yacht Company.** Captain-run yacht management.